Model	Active Params	MMLU	HellaS	WinoG	PIQA	Arc-e	Arc-c	NQ	TriQA	HumanE	MBPP	Math	GSM8K
LLaMA 2 7B	7B	44.4%	77.1%	69.5%	77.9%	68.7%	43.2%	17.5%	56.6%	11.6%	26.1%	3.9%	16.0%
LLaMA 2 13B	13B	55.6%	80.7%	72.9%	80.8%	75.2%	48.8%	16.7%	64.0%	18.9%	35.4%	6.0%	34.3%
LLaMA 1 33B	33B	56.8%	83.7%	76.2%	82.2%	79.6%	54.4%	24.1%	68.5%	25.0%	40.9%	8.4%	44.1%
LLaMA 2 70B	70B	69.9%	**85.4%**	**80.4%**	82.6%	79.9%	56.5%	25.4%	**73.0%**	29.3%	49.8%	13.8%	69.6%
Mistral 7B	7B	62.5%	81.0%	74.2%	82.2%	80.5%	54.9%	23.2%	62.5%	26.2%	50.2%	12.7%	50.0%
Mixtral 8x7B	13B	**70.6%**	84.4%	77.2%	**83.6%**	**83.1%**	**59.7%**	**30.6%**	71.5%	**40.2%**	**60.7%**	**28.4%**	**74.4%**

Table 2: Comparison of Mixtral with Llama. Mixtral outperforms or matches Llama 2 70B performance on almost all popular benchmarks while using 5x fewer active parameters during inference.

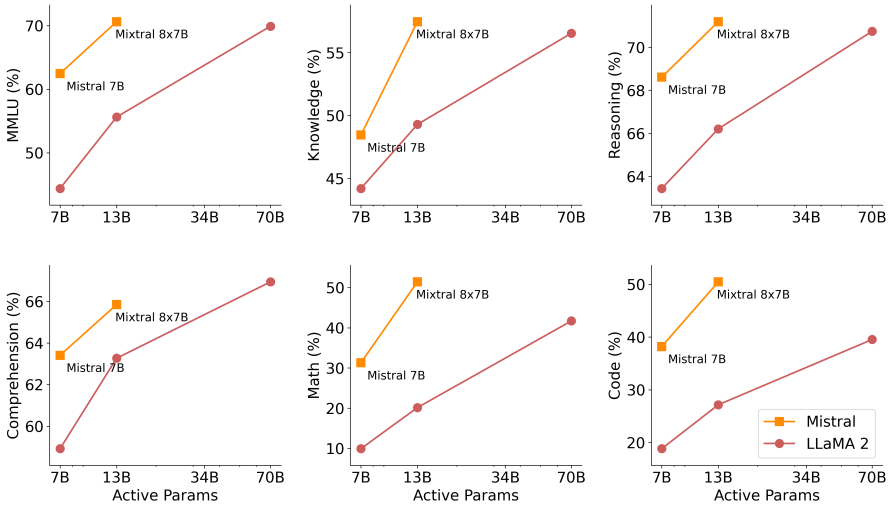


Figure 3: Results on MMLU, commonsense reasoning, world knowledge and reading comprehension, math and code for Mistral (7B/8x7B) vs Llama 2 (7B/13B/70B). Mixtral largely outperforms Llama 2 70B on all benchmarks, except on reading comprehension benchmarks while using 5x lower active parameters. It is also vastly superior to Llama 2 70B on code and math.

Detailed results for Mixtral, Mistral 7B and Llama 2 7B/13B/70B and Llama 1 34B[2] are reported in Table 2. Figure 2 compares the performance of Mixtral with the Llama models in different categories. Mixtral surpasses Llama 2 70B across most metrics. In particular, Mixtral displays a superior performance in code and mathematics benchmarks.

Size and Efficiency. We compare our performance to the Llama 2 family, aiming to understand Mixtral models' efficiency in the cost-performance spectrum (see Figure 3). As a sparse Mixture-of-Experts model, Mixtral only uses 13B active parameters for each token. With 5x lower active parameters, Mixtral is able to outperform Llama 2 70B across most categories.

Note that this analysis focuses on the active parameter count (see Section 2.1), which is directly proportional to the inference compute cost, but does not consider the memory costs and hardware utilization. The memory costs for serving Mixtral are proportional to its *sparse* parameter count, 47B, which is still smaller than Llama 2 70B. As for device utilization, we note that the SMoEs layer introduces additional overhead due to the routing mechanism and due to the increased memory loads when running more than one expert per device. They are more suitable for batched workloads where one can reach a good degree of arithmetic intensity.

Comparison with Llama 2 70B and GPT-3.5. In Table 3, we report the performance of Mixtral 8x7B compared to Llama 2 70B and GPT-3.5. We observe that Mixtral performs similarly or above the two other models. On MMLU, Mixtral obtains a better performance, despite its significantly smaller capacity (47B tokens compared to 70B). For MT Bench, we report the performance of the latest GPT-3.5-Turbo model available, `gpt-3.5-turbo-1106`.

[2]Since Llama 2 34B was not open-sourced, we report results for Llama 1 34B.